

PHYSICAL SCIENCE FOR EDUCATORS VOLUME 1



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Physical Science for Educators Volume 1

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CHAPTER OVERVIEW

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- 1.2: Chemistry and Physics in Context
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- 1.4: Limitations of the Scientific Method
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This lesson is based on Chapter 1 of the OpenStax College Physics textbook. Each section of the chapter will include my lecture slides as well as an embedded link to my YouTube video for that lecture.

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1.1: Introduction and Learning Objectives

Chapter Objectives

Learning Objectives

- **Explore Chemistry in Context:** Analyze the historical development of chemistry, from its early practices in alchemy to its current role as a central science in various fields, highlighting the interconnectedness of chemistry with other STEM disciplines.
- **Understanding the Scientific Method:** Recognize the key components of the scientific method, including observation, hypothesis formulation, experimentation, and analysis, and how they contribute to scientific inquiry.
- **Differentiating Observations:** Distinguish between qualitative and quantitative observations, understanding their roles in forming hypotheses and conducting experiments.
- **Formulating Hypotheses:** Develop the skill to create testable hypotheses based on observations, understanding their importance as a fundamental step in the scientific method.
- **Designing and Conducting Experiments:** Learn to design controlled experiments to test hypotheses, focusing on the manipulation of independent variables and measurement of dependent variables.
- **Interpreting Experimental Results:** Develop the ability to analyze experimental data, determining whether to accept, reject, or modify hypotheses based on empirical evidence.
- **Advancing Hypotheses to Theories and Laws:** Understand the process by which repeated experimental validation of hypotheses leads to the development of scientific theories and laws.
- **Evaluating Science in Media:** Critically assess scientific information presented in media, distinguishing between sound science and "junk science," and recognizing the influence of political or commercial interests.
- **Identifying and Avoiding Pseudoscience:** Learn to recognize pseudoscience and understand its distinction from legitimate scientific inquiry, emphasizing the importance of controlled and interpretable experiments.
- **Applying the Scientific Method in Real-World Contexts:** Explore the application of the scientific method in real-world situations, including observational and manipulative experiments across various scientific fields.
- **Scientific Literacy and Critical Thinking:** Promote scientific literacy and critical thinking, empowering educators to teach the scientific method as a tool for understanding and investigating the natural world.

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1.2: Chemistry and Physics in Context

Throughout human history, our relentless curiosity has driven us to understand the world around us, leading to the exploration and cataloging of a vast array of phenomena. From the basic actions of chipping flint in the Stone Age to the complex processes of isolating chemical substances and creating metallic alloys, humans have sought to manipulate matter to better understand and utilize it. This journey of discovery extends from the mundane to the cosmic—from the flight of birds and the colors of flowers to the clusters of galaxies and the fundamental questions of the universe's creation.



1.1.1: This portrayal shows an alchemist's workshop circa 1580. Although alchemy made some useful contributions to how to manipulate matter, it was not scientific by modern standards. (credit: Chemical Heritage Foundation).

A sketch depicts 4 people stirring and handling chemicals. The chemicals are held in a variety of barrels and large cylinders. Several of the containers are being heated over burning embers. A large stove in the laboratory is filled with burning embers. There is also a large chest in the corner that is producing steam.

Physics and chemistry, as pivotal scientific disciplines, delve into these phenomena by explaining the interactions of energy, matter, space, and time. They uncover the surprisingly simple and unified physical laws that underpin the apparent complexity of the universe. For example, the law of conservation of energy connects seemingly unrelated concepts such as food calories, batteries, car engines, and electric vehicles, illustrating how energy transforms from one form to another without loss. This principle is a cornerstone of both physics and chemistry, illuminating everything from the metabolic processes in biology to the energy transformations in chemical reactions and technological devices.

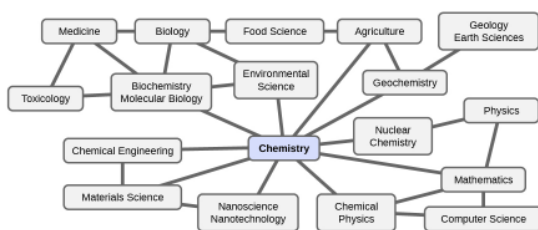


Figure 1.1.2 Knowledge of chemistry is central to understanding a wide range of scientific disciplines. This diagram shows just some of the interrelationships between chemistry and other fields.

A flowchart shows a box containing chemistry at its center. Chemistry is connected to geochemistry, nuclear chemistry, chemical physics, nanoscience and nanotechnology, materials science, chemical engineering, biochemistry and molecular biology, environmental science, agriculture, and mathematics. Each of these disciplines is further connected to other related fields including medicine, biology, food science, geology, earth sciences, toxicology, physics, and computer science.

The role of chemistry in understanding the material composition of the universe complements the principles of physics. Chemistry, often termed the "central science," bridges the physical sciences with life sciences and applied sciences such as medicine and engineering. The interplay between chemistry and physics is evident in fields such as biochemistry and chemical engineering, where the behavior of atoms and molecules is crucial for designing new materials and drugs.

In the laboratory, early alchemists mixed substances, hoping to transform lead into gold, unaware that they were laying the groundwork for modern chemistry. Similarly, physicists from ancient times to the present have sought to understand the forces that

govern the motion of the planets and the properties of light and gravity that influence our everyday lives. This foundational knowledge helps engineers design smartphones and GPS systems that rely on precise interactions of electrical circuits and the laws of motion and relativity.

Science thrives on this interconnectedness, where a discovery in one field can illuminate concepts in another. The shared methodologies and conceptual frameworks of physics and chemistry enable us to solve complex problems and innovate solutions that traverse multiple disciplines. Whether analyzing the chemical properties of new material or studying the forces that bind the cosmos, scientists rely on an integrated approach that respects the underlying simplicity and order of nature.

Understanding these connections enhances our appreciation of the natural world and equips us with the analytical skills necessary for a wide range of careers. By applying the laws of physics and chemistry, we not only decipher how the universe operates but also learn to shape it to improve our quality of life. This educational journey empowers us with a comprehensive toolkit to explore, describe, and manipulate the world around us, ensuring that we continue to innovate and advance our understanding of the universe.

In conclusion, the quest for knowledge in chemistry and physics exemplifies the human desire to seek order and understand the underlying principles of the universe. As we continue to explore these fundamental sciences, we unlock the mysteries of the material world and harness its powers to foster technological advances and improve our everyday lives. This integrated approach in science education highlights the importance of understanding the interrelationships among various scientific disciplines and the real-world applications that these relationships enable.

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1.3: Using the Scientific Method

Learning Objectives

- To identify the components of the scientific method
- Classify measurements as being quantitative or qualitative.
- Evaluate science in the media.

The Scientific Method

Scientists search for answers to questions and solutions to problems by using a procedure called the **scientific method**. This procedure consists of making observations, formulating hypotheses, and designing experiments, which in turn lead to additional observations, hypotheses, and experiments in repeated cycles (Figure 1.3.1).

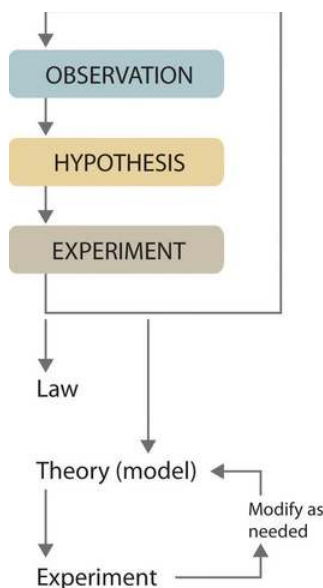


Figure 1.3.1: The Steps in the Scientific Method. (CC BY-SA-NC; anonymous).

Step 1: Make observations

Observations can be qualitative or quantitative. **Qualitative observations** describe properties or occurrences in ways that do not rely on numbers. Examples of qualitative observations include the following: the outside air temperature is cooler during the winter season, table salt is a crystalline solid, sulfur crystals are yellow, and dissolving a penny in dilute nitric acid forms a blue solution and a brown gas. **Quantitative observations** are measurements, which by definition consist of both a number and a unit. Examples of quantitative observations include the following: the melting point of crystalline sulfur is 115.21° Celsius, and 35.9 grams of table salt—whose chemical name is sodium chloride—dissolves in 100 grams of water at 20° Celsius. For the question of the dinosaurs' extinction, the initial observation was quantitative: iridium concentrations in sediments dating to 66 million years ago were 20–160 times higher than normal.

Step 2: Formulate a hypothesis

After deciding to learn more about an observation or a set of observations, scientists generally begin an investigation by forming a **hypothesis**, a tentative explanation for the observation(s). The hypothesis may not be correct, but it puts the scientist's understanding of the system being studied into a form that can be tested. For example, the observation that we experience alternating periods of light and darkness corresponding to observed movements of the sun, moon, clouds, and shadows is consistent with either of two hypotheses:

- a. Earth rotates on its axis every 24 hours, alternately exposing one side to the sun, or
- b. the sun revolves around Earth every 24 hours.

Suitable experiments can be designed to choose between these two alternatives. For the disappearance of the dinosaurs, the hypothesis was that the impact of a large extraterrestrial object caused their extinction. Unfortunately (or perhaps, fortunately), this hypothesis does not lend itself to direct testing by any obvious experiment, but scientists can collect additional data that either support or refute it.

Step 3: Design and perform experiments

After developing a hypothesis, scientists design and conduct experiments to test its validity. **Experiments** are systematic methods of making observations or measurements under controlled conditions, ideally with only one variable altered at a time. This approach allows researchers to isolate the effects of one variable on another in a clear, structured environment.

In any experiment, two primary types of variables are involved:

Independent Variable: This is the variable that the scientist changes deliberately to observe its effect. It's the cause part of cause and effect.

Dependent Variable: This variable is what the scientist measures or observes. It's the effect that is potentially influenced by changes in the independent variable.

Experiments typically have at least two groups:

Control Group: This group does not receive the experimental treatment or change in the independent variable. It serves as a baseline to compare the effects of the independent variable. Some experiments include things called **positive controls** and **negative controls**. These are slightly different from control groups. Positive and negative controls serve to show that the experiment is working correctly. A positive control is a part of the experiment that is deliberately designed to give a positive result. It shows that the experiment is capable of producing a positive result when it is supposed to. A negative control is a part of the experiment that is designed to give a negative result. It shows that the experiment is capable of producing a negative result when it is supposed to.

Treatment Group(s): In these groups, the independent variable is manipulated to various degrees, allowing the scientist to observe the changes that occur in the dependent variable.

After the experiment, scientists collect data, often quantifying it in graphs or tables for analysis. This data analysis helps determine if there is a statistical relationship between the dependent and independent variables, thereby supporting or refuting the hypothesis.

Step 4: Accept or modify the hypothesis

A properly designed and executed experiment enables a scientist to determine whether the original hypothesis is valid. In which case he can proceed to step 5. In other cases, experiments often demonstrate that the hypothesis is incorrect or that it must be modified thus requiring further experimentation.

Step 5: Development into law and/or theory

More experimental data are then collected and analyzed, at which point a scientist may begin to think that the results are sufficiently reproducible (i.e., dependable) to merit being summarized in **law**, a verbal or mathematical description of a phenomenon that allows for general predictions. A law simply says what happens; it does not address the question of why.

One example of a law, the law of definite proportions, which was discovered by the French scientist Joseph Proust (1754–1826), states that a chemical substance always contains the same proportions of elements by mass. Thus, sodium chloride (table salt) always contains the same proportion by mass of sodium to chlorine, in this case, 39.34% sodium and 60.66% chlorine by mass, and sucrose (table sugar) is always 42.11% carbon, 6.48% hydrogen, and 51.41% oxygen by mass.

Whereas a law states only what happens, a **theory** attempts to explain why nature behaves as it does. Laws are unlikely to change greatly over time unless a major experimental error is discovered. In contrast, a theory, by definition, is incomplete and imperfect, evolving with time to explain new facts as they are discovered.

Because scientists can enter the cycle shown in Figure 1.3.1 at any point, the actual application of the scientific method to different topics can take many different forms. For example, a scientist may start with a hypothesis formed by reading about work done by others in the field, rather than by making direct observations.

A Real-World Application of the Scientific Method

In 2007, my husband and I journeyed to China to adopt our daughter. Upon arrival in Beijing, I became violently ill. Due to her visa paperwork, my husband, daughter, and I were required to stay in China for two weeks. Unfortunately, I was ill the entire time. Once the two-week period was up, the three of us flew back to the United States where I continued to be sick.

For the next year, I remained ill and lost a total of 30 pounds. The picture below shows me holding my daughter eight months after we returned home from China.



I would like you to attempt to perform the scientific method on my situation described above. List the steps of the scientific method along with some plausible explanations. * Please have this ready to discuss in class.*

✓ Exercise (PageIndex{1})

Classify each statement as a law, a theory, an experiment, a hypothesis, a qualitative observation, or a quantitative observation.

- Ice always floats on liquid water.
- Birds evolved from dinosaurs.
- According to Albert Einstein, $\text{mass} \times \text{speed of light} = \text{energy}$
- When 10 g of ice was added to 100 mL of water at 25°C, the temperature of the water decreased to 15.5°C after the ice melted.
- The ingredients of Ivory soap were analyzed to see whether it really is 99.44% pure, as advertised.

Solution

- This is a general statement of a relationship between the properties of liquid and solid water, so it is a law.
- This is an educated guess regarding the origin of birds, so it is a hypothesis.
- This is a theory that explains an explanation of events and can be disproven at any time.
- The temperature is measured before and after a change is made in a system, so these are quantitative observations.
- This is an analysis designed to test a hypothesis (in this case, the manufacturer's claim of purity), so it is an experiment.

? Exercise (PageIndex{2})

Classify each statement as a law, a theory, an experiment, a hypothesis, a qualitative observation, or a quantitative observation.

- Measured amounts of acid were added to a Rolaids tablet to see whether it really "consumes 47 times its weight in excess stomach acid."
- Heat always flows from hot objects to cooler ones, not in the opposite direction.
- The universe was formed by a massive explosion that propelled matter into a vacuum.
- Michael Jordan is the greatest pure shooter ever to play professional basketball.
- Limestone is relatively insoluble in water but dissolves readily in dilute acid with the evolution of a gas.
- Gas mixtures that contain more than 4% hydrogen in air are potentially explosive.

Answer a: experiment

Answer b: law

Answer c: theory

Answer d: hypothesis

Answer e: qualitative observation

Answer f: quantitative observation

? Exercise 1.3.1

Classify each statement as a law, a theory, an experiment, a hypothesis, a qualitative observation, or a quantitative observation.

- When 10 g of ice were added to 100 mL of water at 25 °C, the temperature of the water decreased to 15.5 °C after the ice melted.
- Litmus paper dipped in lemon juice turns red.
- A prism separates white light into a spectrum of colors.
- Limestone is relatively insoluble in water but dissolves readily in dilute acid with the evolution of a gas.
- Gas mixtures that contain more than 4% hydrogen in air are potentially explosive.

Answer a: quantitative observation

Answer b: qualitative observation

Answer c: qualitative observation

Answer d: qualitative observation

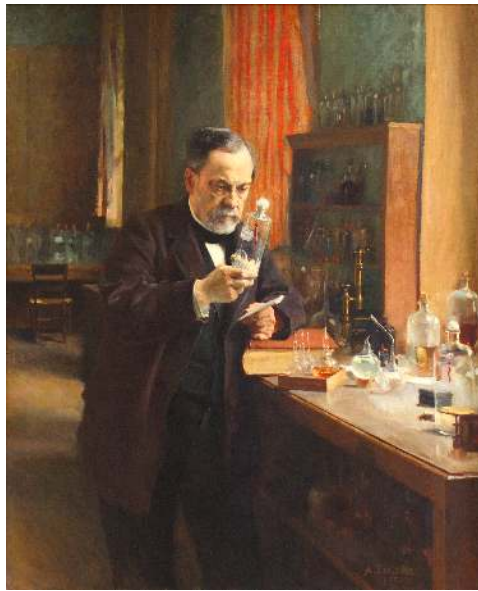
Answer e: quantitative observation

[The Scientific Methods: Crash Course History of Science #14](#)

One famous example is the experiment conducted by Louis Pasteur to test the hypothesis of **spontaneous generation**. This was the once popular idea that non-living material can spontaneously transform into living organisms. For example, if meat were left out too long, people would find that it was infested with maggots. Many people assumed that the meat was transforming into live maggots. After the discovery of microorganisms like bacteria, people questioned where they came from. Were they being spontaneously generated or were they reproducing? In other words, did microorganisms produce new microorganisms?

To test the hypothesis of spontaneous generation, Pasteur devised a simple but ingenious experiment. He filled two flasks with chicken broth. One was an open-necked flask, and the other was a swan-necked flask. The open-necked flask would allow bacteria from the air to enter the broth. On the other hand, the swan-necked flask would trap any bacteria in the elbow of the neck, thus preventing the bacteria from reaching the broth. He then boiled both broths to sterilize them. If spontaneous generation was valid, it should not have mattered whether bacteria from the outside were able to enter the broth. There should be bacteria growing in both flasks.

On the other hand, if bacteria only came from the outside, then only the open-necked flask should have bacteria growing in it. After a few days, Pasteur examined the broth in both flasks and found only new bacteria growing in the open-necked flask. Since the swan-neck flask did not show any bacteria growing in it, the hypothesis of spontaneous generation was rejected, and we now know that for maggots to grow on meat, flies must first lay their eggs on the meat. The maggots then hatch out of the eggs and feed on the rotting meat. The meat itself did not spontaneously generate the maggots.



Albert_Edelfelt_-_Louis_Pasteur_-_1885.jpg Photo from Wikimedia Commons

To put Pasteur's experiment into the above terms, the shape of the flask's neck was the independent variable and the amount of bacterial growing in each flask was the dependent variable that Pasteur tested for. The open-neck flask was the control group used to compare the swan-neck flask that was used for the treatment group.



Wikimedia commons L0057281 Copy of Pasteur's flask used in his experiments on spontaneous generation Credit: Science Museum, London. Wellcome Images images@wellcome.ac.uk http://wellcomeimages.org/Copy_of_Pas...e_L0057281.jpg



Erlenmyer flask By Hannes Grobe 19:04, 3 September 2006 (UTC) - Own work, CC BY-SA
2.5,<https://commons.wikimedia.org/w/index.php>

The above is an example of a **manipulative experiment** in which a single variable is to be changed under controlled conditions. Many people think that manipulative experiments are the only way in which science is done, but there are many cases in which the phenomena being studied are too big or too distant in space or time to be studied by a controlled experiment in the lab. In these cases, scientists learn by doing **observational science**. For example, we cannot build a star in a lab and study it up close. Fortunately, our galaxy is filled with billions of stars of different sizes, ages, and temperatures. By collecting data from many stars in various stages of stellar evolution, astronomers can build a model for how stars form and change over time. Astronomy is, therefore, largely an observational science.

A scientific experiment must be repeatable, that is, you or someone else conducting the same study should get similar results. One case might be a fluke, but if several repeats of the experiment yield similar results, then the hypothesis is better supported.

Once the scientist has completed their study, the usual practice is for them to write a paper and submit it to a peer-reviewed journal. Peer review is the process by which scientists in the same field evaluate each other's work. When a journal receives a proposed paper, it sends it out to several other scientists (the "peers") who review it and recommend whether to publish it. If the peers conclude that the research followed good scientific methodology and the data support the conclusions, they recommend publication. Otherwise, the paper is rejected, and the scientist has to do more work before resubmitting.

Unfortunately, there are many places where "scientific" papers can be published without peer review. Many of these are pay-for-publishing journals that will publish almost any paper if the researcher pays a fee. Also, some organizations with a political agenda may self-publish what looks like legitimate research that is slanted to reach a predetermined conclusion.

Today, few people dispute the fact that there is a strong link between smoking tobacco and certain forms of cancer. However, for decades, doubt was sown by an organization called the Tobacco Institute, which published many convincing-looking papers. All these papers came to the same conclusion: "Gosh, we just can't find any link between tobacco and cancer." This was the opposite of what nearly every other researcher in the field concluded. However, to be able to come to different results? Well, the Tobacco Institute was funded by the tobacco industry, and we now know that they were under orders by their sponsors to come to predetermined conclusions no matter what the data said.

More recently, a research paper in a pay-to-publish journal created a sensation in the news media by announcing that you can lose weight by eating chocolate. Wouldn't that be great? Sounds too good to be true, right? Well, it was. The researcher behind it came forward to admit that the entire paper was a hoax. He did it to highlight how easy it is to get bogus studies published and how readily the media can hype what seem like sensational results. Sadly, most journalists are not trained to discern the difference between sound science and what we can label **"junk science"** or **pseudoscience**.

Science plays a big role in our understanding of the world around us. It is, therefore, imperative that we have a scientifically literate society that is capable of discerning good science from junk science, especially when making decisions relating to their health and well-being.

One final word on science and relates to how people use the word **theory**. Many non-scientists use the word to mean a guess or a hunch based on incomplete information. However, this is now how scientists use the term. In science, a theory is a broad

explanation for a phenomenon that has been well-tested, shown to be supported repeatedly by experiments, and has gained wide acceptance. It is not a single hunch or guess. The way non-scientists use the term theory is more akin to how scientists use the word hypothesis. Another misconception is that if a hypothesis is validated by experiment, it may be “promoted” to become a theory. That is not accurate either. A **theory** is a broader explanation, while a hypothesis is generally narrower in scope. Indeed, a single theory may encompass several hypotheses. A theory offers a deeper, more detailed explanation for why things happen as they do in nature, like Einstein's Theory of Relativity explaining gravity through the curvature of space-time. It's not just an isolated observation; it's a comprehensive model that explains a wide range of phenomena.

On the other hand, a law describes what happens based on consistent observational data, such as Newton's Laws of Motion that describe how objects move but do not explain why they move that way. Laws provide a description of an observed phenomenon without looking into the mechanics behind it. They are universally accepted as true under consistent conditions and help predict outcomes.

It's important to note that in science, a hypothesis does not simply "evolve" into a theory through validation. A theory is a broader and more complex explanation that often encompasses several hypotheses. The transition from hypothesis to theory involves a significant accumulation of evidence and broader acceptance in the scientific community.

Keep that in mind when someone dismisses a scientific principle as “just a theory.” Nearly everything you are taught in a science class is based on theory. Of course, just because a theory has gained broad acceptance does not always mean it is true, but it takes a lot of evidence to overturn a theory that has been repeatedly validated by observation. Thus, much of what you learn in science classes is based on theories—well-established, widely tested, and crucial for advancing our understanding of the natural world.

Evaluating Science in the Media



Contributors and Attributions

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1.4: Limitations of the Scientific Method

Learning Objectives

- To understand the limitations in the scientific method, one must become familiar with the scientific method and its components.

Pseudo-science, basically "fake"-science," consists of scientific claims which are made to appear factual when they are actually false. Many people question whether Pseudo-science should even contain the word "science" as Pseudo-science isn't really even an imitation of science; it pretty much disregards the scientific method all together. Also known as alternative or fringe-science, Pseudo-science relies on invalid arguments called sophisms, a word Webster dictionary defines as "an argument apparently correct in form but actually invalid; especially : such an argument used to deceive". Pseudo-science usually lacks supporting evidence and does not abide by the scientific method. That is, pseudo-theories fail to use carefully cultivated and controlled experiments to test a hypothesis. A scientific hypothesis must include observable, empirical and testable data, and must allow other experts to test the hypothesis. Pseudo-science does not accomplish these goals. Several examples of Pseudo-Science include phrenology, astrology, homeopathy, reflexology and iridology.

Distinguishing Pseudo-Science

In order to distinguish a pseudoscience, one must look at the definition of science, and the aspects that make science what it is. Science is a process based on observations, conjectures, and assessments to provide better understanding of the natural phenomena of the world. Science generally always follows a formal system of inquiry which consists of observations, explanations, experiments, and lastly, hypothesis and predictions. Scientific theories are always challenged by experts and revised to fit new theories. Pseudo-science, however, is mostly based on beliefs and it greatly opposes contradictions. Their hypothesis are never revised to fit new data or information. Scientist continually disprove ideas to achieve a better understanding of the physical world, whereas pseudo-scientists focus on proving theories to make their claims seem plausible. For example, science text books come out with new editions every couple of years to correct typos, update information, add new illustrations, etc. However, it has been observed that pseudo-science textbooks only come out with one edition, and is never updated or revised even if their theory has been proven to be false.

Pseudo-science beliefs often tend to be greatly exaggerated and very vague. Complicated technical language is often used to sound impressive but it is usually meaningless. For example, a phrase like "energy vibrations" is used to sound remarkable but a phrase like this is insignificant and doesn't really explain anything. Furthermore, Pseudo-science often consists of outrageous, yet unprovable claims. Thus, pseudo-scientists tend to focus on confirming their ideas, rather than finding evidence that refutes them. The following dialogue contains the thought-processes behind Pseudo-Science.

1. My friend and I think unicorns exist
2. Science has no evidence about unicorns.
3. Science therefore cannot prove if unicorns do or do not exist.
4. One day my friend, a very trustworthy person, said she saw a unicorn in the field by her house. There is no other evidence, other than the fact that my friend saw it.
5. Unicorns exist and any scientist who tries to deny the existence of unicorns is a fun-sucking, hostile human being.

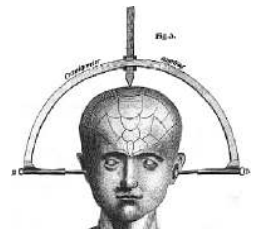
The dialogue above features many key characteristics of Pseudo-Science. The speaker makes his or her point valid though the two facts alone that her friend had a personal experience and that science has no proof to prove the theory wrong. Finally, the speaker insults anyone who would challenge the theory. In science, challenges to a theory are accepted as everyone has the same common goal of improving the understanding of the natural world. Below is a table that lays out the key characteristics of Science and Pseudo-Science

SCIENCE	PSEUDO-SCIENCE
Science never proves anything.	Pseudoscience aims to prove an idea.
Self-correcting methodology which involves critical thinking.	Starts with a conclusion and gives easy answers to complex problems.

SCIENCE	PSEUDO-SCIENCE
An on-going process to develop a better understanding of the physical world by testing all possible hypotheses.	Often driven by social, political or commercial goals.
Involves a continual expansion of knowledge due to intense research.	A field has not evolved a lot since the beginning. If any research is done, it is done to justify the claims, rather than expand them.
Scientists constantly attempt to refute other scientists' works.	An attempt to disprove the beliefs is considered hostile and unacceptable.
When results or observations are not consistent with a scientific understanding, intense research follows.	Results or observations that are not consistent with current beliefs are ignored.
Remains questionable at any time. There are two types of theories: those that have been proven wrong by experimentation and data, and every other theory. Thus, no theory can be proven correct; every theory is also subject to being refuted.	Beliefs of the field can not usually be tested empirically so will likely not ever be proven wrong; Thus, Pseudo-scientists believe that they are right just because no one can prove them wrong.
Concepts are based on previous understandings or knowledge.	Pseudo-Scientists are often not in touch with main-stream science and are often driven by the egos of the "scientists". Furthermore, famous names and testimonials are often used for support rather than scientific evidence.
Findings must be stated in unambiguous, clear language.	Pseudoscience often uses very vague, yet seemingly technical terms.

Phrenology

Phrenology, also known as craniology, was a "science" popular during the early 1800s that was centered around the idea that the brain was an organ of the mind. During this time, most people believed that the brain was divided into distinct sections that all controlled different parts of a person's personality or intelligence. The basis of phrenology revolves around the concept that the brain mirrors a muscle and those parts of the brain which are "exercised" the most, will be proportionally larger than those parts of the brain that aren't often used. Thus, the scientists pictured the brain as a bumpy surface, with the make-up of the surface differing for every person depending on their personality and intelligence. By the mid 19th century, automated phrenology machines existed, which was basically a set of spring loaded probes that were placed on the head to measure the topography of one's skull. The machine then gave an automated reading about a person's characteristics based on this.



Let's consider some of the key characteristics of pseudo-science from our chart, and see how they apply to phrenology.

- Pseudo-Scientists are often not in touch with main-stream science: Scientific research has since the 1800s shown how though the brain is indeed divided into sections, each section does not determine a characteristic or personality trait, but instead controls a specific function such as memory or motor skills. Likewise, it has been concluded that the brain conforms to the shape of the skull, rather than the skull conforming to the shape of the brain (meaning the bumps of a person's skull have nothing to do with the shape of the brain). Back in the 1800s, little knowledge existed about the realities of brain structure and function, so the concept wasn't as reflexive of pseudo-science as it is today. However, some doctors and scientist still believe in the basic tenets of phrenology. Phrenology today exists as a classic form of pseudo-science as it goes against the common understanding about how the brain functions.
- Often driven by social, political or commercial goals- Indeed, the main goal of phrenology was a political and social one: to prove the dominance of the white race over other races. "Scientists" measured the brains of both races and concluded that the brains of white people were larger than that of people of African descent. Therefore, they concluded, they were smarter and superior. It was later revealed that the scientists were biased while conducting the experiment and that they were previously aware of what race each brain belonged to. The experiment was repeated and this time the scientists were not aware of the race and they concluded that the brains were of equal size. The second experiment better conforms to the scientific method, as in this case the scientists objectively measured the brains, while in the first case the bias of the scientists led to their conclusions. Thus, this situation demonstrates a two-fold level of defective science because not

only was the idea of measuring the brains to determine personality and intelligence not correct all together, but the methods in which the scientists were doing this was also flawed. Phrenology was also commercially driven, since phrenology parlors were very wide spread and many devices were on the market to be used to measure.

- Pseudo-Scientists are often driven by the egos of the "scientists"- In the book *Phrenology and the origins of Victorian Scientific Naturalism* by John Van Whye, Van Whye quotes about the main discoverer of Phrenology Franz Joseph Gall, that "the peculiar incentive behind Gall's fascination with explaining individuals' differences may have lain in his hubris" (Van Whye 18). Of the 12 children in his family, Gall was the sharpest and brightest and naturally interested in distinguishing factors between children. Even as a young school boy, Gall noticed that the other children who were just as good at memorization as he was all had protruding eyes, which lead him to the idea of the basis of phrenology, that the characteristics of one's head indicates his or her intelligence.

Reflexology

Reflexology is a way of treatment that involves physically applying pressure to the feet or hands with the belief that each are divided up into different zones that are "connected" to other parts of the body. Thus, reflexologists assert that they can make physical changes throughout the body simply by rubbing ones hands or feet. Like we did with phrenology, lets go through some of the main characteristics of Pseudo-Science and see how they apply to reflexology.

- Pseudo-Scientists are often not in touch with main-stream science: No Scientific research has proven the validity of reflexology and how in fact it would actually work. In 2009, the Australian Medical Journal conducted an extensive study on reflexology and concluded "The best evidence available to date does not demonstrate convincingly that reflexology is an effective treatment for any medical condition". However, despite this lack of evidence, Reflexology continues.
- Pseudoscience often uses very vague, yet seemingly technical terms A main focus of reflexology is that the pressure on the foot removes any blockage of Qi, the "life energy force" and restores balance to lead to better health. Terms like "vital energy" or "energy blockage" which are used to talk about reflexology are classic pseudo-science terms; they sound impressive yet have no meaning to us
- Furthermore, famous names and testimonials are often used for support rather than scientific evidence. Because pseudo-science beliefs do not use scientific data for support, they must rely on individual circumstances when their product, idea, etc. appeared to have worked. For example, on the home page of well-known reflexologist Laura Norman's home page, she has a quote of Regis Philbin (past host of *Who Wants to be a Millionaire?*) saying "Laura Norman's Reflexology spared me from a kidney stone operation and saved my life.", opposed to a quote from say, a medical journal, that would cite how many studies say reflexology is an extremely effective form of treatment.

Distinguishing Pseudo-Science from other types of invalid science

An important distinction should be made between Pseudo-science and other types of defective science. Take for example, the "discovery" of N-rays. While attempting to polarize X-rays, physicist René Prosper Blondlot claimed to have discovered a new type of radiation he called N-rays. After Blondlot shared with others his exciting discovery, many other scientists confirmed his beliefs by saying they too had saw the N-rays. Though he claimed N-rays contained impossible properties, Blondlot asserted when he put a hot wire in an iron tube, he was able to detect the N-rays when he used a thread of calcium sulfite that glowed slightly when the rays were sent through a prism of aluminum. Blondlot claimed that all substances except some treated metals and green wood emit N-rays. However, Nature magazine was skeptical of Blondlot and sent physicist Robert Wood to investigate. Before Blondlot was about to show Wood the rays, Wood removed the aluminum prism from the machine without telling Blondlot. Without the prism, the rays would be impossible to detect. However, Blondlot claimed to still see the N-rays, demonstrating how the N-rays did not exist; Blondlot just wanted them to exist. This is an example of Pathological science, a phenomenon which occurs when scientists practice wishful data interpretation and come up with results they want to see. This case of Pathological science and Pseudo-science differ. For one, Blondlot asked for a confirmation by other experts, something Pseudo-science usually lacks. More importantly, in pathological science, a scientist starts by following the scientific method; Blondlot was indeed doing an experiment when he made his discovery and proceeded to experiment when he found the substances that did not emit the rays. However, Pseudo-science usually includes a complete disregard of the scientific method, while Pathological scientists includes following the scientific method but seeing the results you wish to see.

Another type of invalid science, called hoax science occurred in 1999 when a team at the Lawrence Berkeley National Laboratory claimed to have discovered elements 116 and 118 when they bombarded Lead with Krypton particles. However, by 2002 it had been discovered that physicist Victor Ninov had intentionally fudged the data to get the ideal results. Thus, the concept of hoax

science, which occurs when the data is intentionally falsified, differs both from pathological and pseudo science. In pathological science, scientists wishfully interpret the data and legitimately think they see what they want to see. However, in Hoax science, scientists know they don't see what they want to see, but just say they did. Finally, in Pseudo-Science, scientists don't consider the scientific method at all as they don't use valid experiments to back up their data in the first place.

From Pseudo-Science to Science

There have been incidents where what was once considered pseudo-science became a respectable theory. In 1911, German astronomer and meteorologist Alfred Wegener first began developing the idea of Continental Drift. The observation that the coastlines of African and South American seemed to fit together was not a new observation: scientists just couldn't believe that the continents could have drifted so far to cross the 5,000 mile Atlantic Ocean. At the time, it was a common theory that a land bridge had existed between Africa and Brazil. However, one day in the library Wegener read a study about a certain species that could not have crossed the ocean, yet had fossils appeared on both sides of the supposed land bridge. This piece of evidence lead Wegener to believe that our world had once been one piece, and had since drifted apart. However, Wegener's theory encountered much hostility and disbelief. In this time, it was the norm for scientists to stay within the scopes of their fields, meaning biologists did not study physics, chemists did not study oceanology and of course, meteorologists/astronomers like Wegener did not study geology. Thus, Wegener's theory faced much criticism just due to the fact that he was not a geologist. Also, Wegener could not explain why the continents moved, just that they did. This lack of reasoning lead to more skepticism about the theory and all these factors combined lead to the viewing of continental drift as Pseudo-Science. However, today much evidence exists that shows that Continental Drift is a perfectly acceptable scientific theory. Today, the modern ideas of plate tectonics can help explain Continental Drift, as the Plate Tectonic Theory presents the idea that the earth's surface is made up of several large plates that often move up to a few inches every year. Also, the development of paleomagnetism, which allows us to determine the earth's magnetic poles at the time a rock formed, suggests that the earth's magnetic poles have changed many times in the last 175 million years and that at one time South America and Africa were connected.

Limitations of the Scientific Method

Due to the need to have completely controlled experiments to test a hypothesis, science can not prove everything. For example, ideas about God and other supernatural beings can never be confirmed or denied, as no experiment exists that could test their presence. Supporters of Intelligent Design attempt to convey their beliefs as scientific, but nonetheless the scientific method can never prove this. Science is meant to give us a better understanding of the mysteries of the the natural world, by refuting previous hypotheses, and the existence of supernatural beings lies outside of science all together. Another limitation of the scientific method is when it comes making judgements about whether certain scientific phenomena are "good" or "bad". For example, the scientific method cannot alone say that global warming is bad or harmful to the world, as it can only study the objective causes and consequences. Furthermore, science cannot answer questions about morality, as scientific results lay out of the scope of cultural, religious and social influences.

Concept Assessment

Determine if each statement is true or false (see answers at bottom of the page)

1. What is considered Pseudo-Science today will always be considered Pseudo-Science
2. A person has a cold and decides to seek reflexology treatment. The next day, the person gets better. This means reflexology is a valid scientific theory
3. Just because "science" is immoral or defective does not necessarily mean it is Pseudo-Science
4. Famous people are used in advertisements for products such as gatorade. This means these products are Pseudo-Science
5. Medically based Pseudo-Science such as homeopathy, reflexology or acupuncture have absolutely no benefits to people

References

1. Ernst, Ezard. "Is Reflexology an Effective Intervention? A Systematic Review of Randomised Controlled Trials." *The Medical Journal of Australia* 191.5 (2009): 263-66. Print.
2. Van, Wyhe John. *Phrenology and the Origins of Victorian Scientific Naturalism*. Aldershot, Hanst, England: Ashgate, 2004. Print.
3. Yount, Lisa. *Alfred Wegener: Creator of the Continental Drift Theory*. New York: Chelsea House, 2009. Print.

Answers to concept assessment

1. False- just because something is considered pseudo-science today, does not mean it will always be. Take for example, our discussion about Continental Drift. Continental Drift used to be considered Pseudo-Science, but now since there is scientific evidence to prove it, the theory is considered a product of science.
2. False- Just because a person got better after having reflexology treatment does not mean the treatment, which has no scientific evidence behind it, is the sole reason for a person's recovery. Many other factors could have lead to a person's healing, such as medication or time to let the body fight by itself so it would be impossible to determine that reflexology caused a person to get over a cold
3. True- Pseudo-Science is a specific type of defective. See the discussion about pathological and hoax science to learn how to distinguish Pseudo-Science from other types of invalid Science.
4. False- The common characteristic of relying on testimonials or celebrity support of Pseudo-Science is just one of the many characteristics on Pseudo-Science. Before declaring something as Pseudo-Science or science, it is important to consider various characteristics of both and focus on whether or not the ideas have experimentally determined data to support them. There has indeed been Scientific Data to support the use of Gatorade.
5. False- though there is little scientific evidence to support these types of medical treatment, it does not mean that they have no value. The Placebo effect may be relevant here, as people may believe that the methods are working, which may trigger the body to actually feel better.

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1.5: Teaching the Scientific Method

What is a Lesson Plan?

A lesson plan is a detailed outline of the teaching objectives, materials, activities, and assessment methods for a specific lesson. It serves as a guide for teachers to ensure that instructional goals are met and provides a structured approach to teaching a particular topic. Here's what a typical lesson plan includes:

- **Objectives:** Clear statements about what students will learn and be able to do by the end of the lesson.
- **Materials:** A list of all the supplies and resources needed for the lesson.
- **Procedure:** Step-by-step instructions on how to conduct the lesson, including introduction, main activities, and conclusion.
- **Assessment:** Methods for evaluating whether students have understood the lesson objectives, such as worksheets, quizzes, or class discussions.
- **Extensions:** Additional activities or projects to reinforce the lesson or explore the topic further.

By using a lesson plan, teachers can ensure that they cover all necessary content, engage students effectively, and assess their understanding in a structured and organized manner.

Teaching the Scientific Method to K-5th Graders

The scientific method is a fundamental process in science that helps students understand how scientists investigate and learn about the world. Teaching this method to K-5th graders involves breaking down complex concepts into simpler, engaging activities that foster curiosity and critical thinking. Here's how to effectively introduce the scientific method to young students using a structured lesson plan.

Lesson Plan: Introduction to the Scientific Method

Objective: Students will learn the steps of the scientific method and apply them in a simple, hands-on experiment.

Materials:

- Clear plastic cups
- Water
- Food coloring
- Paper towels
- Small objects (e.g., coins, pebbles)
- Worksheet for recording observations

Steps of the Scientific Method:

1. **Make Observations:** Begin with a discussion about observations. Explain that scientists use their senses to learn about the world around them. Show students a cup of water and ask them to describe what they see (e.g., "The water is clear," "It's in a plastic cup").
2. **Formulate a Hypothesis:** Introduce the concept of a hypothesis as an educated guess about what might happen in an experiment. For example, ask, "What do you think will happen if we add food coloring to the water?" Guide students to form simple hypotheses like, "The water will turn blue."
3. **Design and Perform Experiments:** Explain that experiments test hypotheses. Divide students into small groups and give each group a cup of water, a drop of food coloring, and a paper towel. Instruct them to add a drop of food coloring to the water and observe what happens. Encourage them to record their observations on the worksheet.
4. **Collect Data:** Teach students how to collect and record data. For this experiment, they can draw what the water looks like before and after adding the food coloring.
5. **Analyze Results:** Discuss the results as a class. Did the water change color as expected? Why or why not? Help students understand that their observations either support or refute their hypotheses.
6. **Accept or Modify Hypothesis:** If the results support the hypothesis, explain that this means their guess was correct. If not, discuss what might have gone wrong and how they could change their hypothesis and try again.
7. **Develop into Law or Theory:** For K-5 students, simplify this step by explaining that when scientists see the same results over and over, they can make rules about how things work. For example, "Every time we add blue food coloring to water,

it turns blue.”

Example Activity: Floating and Sinking

Objective: Students will understand the concept of density and apply the scientific method to predict and test which objects will float or sink in water.

Materials:

- Clear plastic bin filled with water
- Various small objects (e.g., coin, cork, plastic toy, pebble)
- Prediction worksheet

Procedure:

1. **Observation:** Show students the objects and the bin of water. Ask them to observe the objects and think about which ones might float or sink.
2. **Hypothesis:** Have each student pick an object and predict whether it will float or sink. Record their predictions on the worksheet.
3. **Experiment:** One by one, place each object in the water. Ask students to observe what happens and record the results.
4. **Data Collection:** Create a chart on the board with two columns: “Float” and “Sink.” Fill in the chart as students share their observations.
5. **Analysis:** Discuss the results. Were their predictions correct? Why did some objects float while others sank? Introduce the concept of density in simple terms.
6. **Conclusion:** Help students understand that their hypotheses can be tested and revised. If an object didn’t behave as predicted, discuss what might have been different about that object (e.g., weight, material).

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1.6: End of Chapter Activity- Develop a Lesson Plan on the Scientific Method

Assignment: Develop a Lesson Plan on the Scientific Method

Objective: Create your lesson plan on the scientific method to teach a different concept or experiment suitable for K-5th grade students.

Assignment:

1. Choose a Topic:

- Select a simple scientific concept or age-appropriate experiment for K-5th grade students.
- Examples: Effects of sunlight on plant growth, how different liquids affect objects or simple weather experiments.

2. Create a Lesson Plan:

◦ **Introduction:**

- Start with a discussion about the chosen topic and why it is interesting or important.

◦ **Explain the Steps of the Scientific Method:**

- Define and explain each step with examples relevant to the chosen topic.

◦ **Plan the Experiment:**

- Detail how the experiment will be conducted, what materials are needed, and how data will be collected.

◦ **Conduct the Experiment:**

- Outline how students will perform the experiment, ensuring it is safe and manageable.

◦ **Data Analysis:**

- Describe how students will analyze the data and what they should look for.

◦ **Conclusion:**

- Plan for students to present their findings and discuss what they learned.

3. Reflection:

- After creating the lesson plan, write a brief reflection on how the scientific method can help students understand and explore scientific concepts.

Developing your own lesson plans, will reinforce your understanding of the scientific method and teach you how to effectively teach this important concept to young students. This hands-on approach not only engages students but also fosters a curiosity and love for science from an early age.

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1.7: End of Chapter Key Terms

Elemental Beginnings: Foundations of Physics and Chemistry Key Terms

1. **Control Group:** A group in an experiment that does not receive the experimental treatment or change in the independent variable, serving as a baseline to compare the effects of the independent variable.
2. **Dependent Variable:** The variable that is measured or observed in an experiment, potentially influenced by changes in the independent variable.
3. **Experiments:** Systematic methods of making observations or measurements under controlled conditions, ideally with only one variable altered at a time.
4. **Hypothesis:** A tentative explanation for an observation or set of observations that can be tested through further investigation.
5. **Independent Variable:** The variable that the scientist deliberately changes to observe its effect, representing the cause in a cause-and-effect relationship.
6. **Junk Science or Pseudoscience:** Research or claims presented as scientific but lacking rigorous standards and peer review, often with predetermined conclusions or biased interpretations.
7. **Law:** A verbal or mathematical description of a phenomenon that allows for general predictions, describing what happens without explaining why.
8. **Manipulative Experiment:** An experiment in which a single variable is deliberately changed under controlled conditions to isolate its effect on another variable.
9. **Negative Controls:** Parts of an experiment designed to give a negative result, showing that the experiment is capable of producing a negative result when it is supposed.
10. **Observational Science:** A scientific approach where phenomena are studied without manipulation, often due to the impracticality of controlled experiments, such as in astronomy.
11. **Observations:** Initial steps in the scientific method, which can be qualitative (describing properties or occurrences without numbers) or quantitative (measurements consisting of numbers and units).
12. **Positive Controls:** Parts of an experiment designed to give a positive result, demonstrating that the experiment is capable of producing a positive result when it is supposed to.
13. **Qualitative Observations:** Descriptions of properties or occurrences that do not rely on numerical measurements, such as the color of sulfur crystals.
14. **Quantitative Observations:** Measurements that consist of both a number and a unit, such as the melting point of a substance.
15. **Scientific Method:** A procedure for searching for answers to questions and solutions to problems through making observations, formulating hypotheses, designing experiments, and collecting data in repeated cycles.
16. **Spontaneous Generation:** An outdated hypothesis suggesting that non-living material can spontaneously transform into living organisms, disproven by experiments such as those conducted by Louis Pasteur.
17. **Theory:** A broad explanation for a phenomenon that has been well-tested, supported by repeated experiments, and widely accepted, explaining why nature behaves as it does.
18. **Treatment Group(s):** Groups in an experiment where the independent variable is manipulated to various degrees to observe the changes that occur in the dependent variable.

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2.1: Introduction and Learning Objectives

Introduction to Units, Measurements, Graphing, and Calculation

Welcome to the fascinating world of physical science, where precision, accuracy, and methodical problem-solving lay the foundation for discovery and innovation. As future educators, you will be guiding young minds through the complexities of scientific principles, fostering their curiosity, and nurturing their understanding of the natural world. This chapter, "Units, Measurements, Graphing, and Calculation," is designed to equip you with the essential mathematical and analytical skills necessary to teach physical science effectively.

We begin with a thorough review of fundamental math concepts, from the order of operations and handling negative numbers to mastering decimals and fractions. These basics form the bedrock of scientific calculations and are crucial for ensuring that students grasp more complex topics down the line. You'll learn how to manipulate formulas, calculate perimeters and circumferences, and work with percentages in various contexts, including error analysis.

Moving forward, we'll delve into geometry, exploring angles, triangles, polygons, and composite figures. You'll gain the ability to calculate areas and volumes of common solids, skills that are indispensable in both classroom demonstrations and real-world applications. Understanding and converting units of measurement is another critical component of this chapter, as it ensures consistency and accuracy in scientific inquiry.

Graphing is another vital skill you'll develop here. Accurate data representation through various types of graphs is not only a fundamental scientific practice but also an engaging way to help students visualize and interpret data. You'll learn how to choose the appropriate graph type for different data sets and ensure clarity and precision in your presentations.

Dimensional analysis, precision, accuracy, and significant figures will round out your mathematical toolkit, enabling you to perform and teach scientific calculations with confidence. By mastering these concepts, you'll be prepared to guide your students through the intricacies of physical science, helping them to develop critical thinking and analytical skills that will serve them well in their academic and professional futures.

This chapter is not just about learning techniques and formulas; it's about building a strong foundation for scientific literacy. As you engage with the material, remember that your role as an educator is to inspire and empower the next generation of scientists, engineers, and informed citizens. Let's embark on this journey together, fostering a deep understanding of the principles that govern our physical world.

Learning Objectives

Chapter 2: Units, Measurements, Graphing, and Calculation

1. Review mathematical concepts.
2. Analyze errors using percentages.
3. Calculate and interpret mean, median, and mode in data sets.
4. Calculate and apply basic probability concepts.
5. Calculate and interpret standard deviation in data sets.
6. Apply the rules of exponents in mathematical calculations.
7. Use scientific notation to express large and small numbers.
8. Differentiate between precision and accuracy in measurements.
9. Apply significant figures in scientific calculations.
10. Perform accurate measurements using various tools.
11. Differentiate between measurement systems and apply them.
12. Use and convert US measurement units in scientific calculations.
13. Apply metric units in scientific calculations.
14. Perform conversions between different measurement systems.
15. Create and interpret graphs to represent data.
16. Choose appropriate graph types for different data sets.
17. Use dimensional analysis for unit conversions.
18. Apply chapter concepts in practical, real-world activities.
19. Review and apply key terms from the chapter in scientific contexts.

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SECTION OVERVIEW

2.2: Math Review

2.2.1: Order of Operations

2.2.2: Negative Numbers

2.2.3: Decimals

2.2.4: Fractions

2.2.5: Formulas

2.2.6: Perimeter and Circumference

2.2.7: Percents Part 1

2.2.8: Ratios, Rates, Proportions

2.2.9: Percents Part 2 and Error Analysis

2.2.10: Percents Part 3

2.2.11: Angles

2.2.12: Triangles

2.2.13: Area of Polygons and Circles

2.2.14: Composite Figures

2.2.15: Surface Area of Common Solids

2.2.16: Converting Units of Area

2.2.17: Volume of Common Solids

2.2.18: Area of Regular Polygons

2.2.19: Pyramids and Cones

2.2.20: Mean, Median, Mode

2.2.21: Probability

2.2.22: Standard Deviation

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